Hidden Economies of Care: Uncredited Women in Early Scientific Networks (1890–1939)

Introduction: Unveiling the Hidden Economies of Scientific Care

The period between 1890 and 1939 witnessed the dramatic professionalization and institutionalization of science. As disciplines solidified and research became increasingly collaborative and resource-intensive, formal systems for assigning credit, primarily through authorship on publications, became paramount. Yet, these nascent credit economies often operated within, and were shaped by, prevailing social and gender norms that rendered certain forms of labor invisible. This report examines the "hidden economies of care" within science during this era – the substantial, yet largely unacknowledged, contributions of women who performed essential tasks underpinning scientific discovery but rarely received formal recognition. Their work, encompassing data organization, complex calculations, collections management, manuscript editing, and myriad other activities, constituted a vital substrate for knowledge production. Recovering and analyzing this labor necessitates moving beyond traditional historical accounts focused solely on published authors and employing methodologies attuned to tracing influence and contribution through less conventional archival traces and analytical frameworks.

This investigation adopts a feminist historiographical approach, which seeks not merely to add women to existing narratives but to critically analyze the gendered structures and assumptions that shaped scientific practice and credit allocation itself. Central to this endeavor is the application of several analytical lenses: a **Citation Ghosting Lens** focuses on reconstructing influence in the absence of authorship by scrutinizing acknowledgments, footnotes, and financial records; **Social Network Analysis (SNA)** provides tools to map relationships and quantify potential contributions obscured in textual accounts; a **Gendered Infrastructure Lens** reframes care and support work as technically essential, foundational labor rather than mere auxiliary help; a **Trans-Contextual Lens** examines how these dynamics played out across different scientific disciplines and institutional settings, from domestic laboratories to colonial botany stations; and a **Present Tense Lens** draws connections between these historical patterns and contemporary issues of uncredited labor in science and technology. Throughout this analysis, a commitment to interpretive caution is maintained, inferring contributions gently from multivalent sources and carefully distinguishing between care, labor, and potential intellectual input, while resisting simplistic "lost genius" narratives.

This report proceeds by first exploring the texture of this invisible work through archival evidence, detailing the specific tasks women performed. It then moves to reconstruct potential influence using methods designed to detect presence without formal authorship, including SNA. Subsequently, it analyzes how this labor functioned as a form of gendered infrastructure, essential yet devalued. The analysis then broadens to consider variations across different scientific fields and institutional contexts. Finally, the report connects these historical patterns to contemporary issues of credit and recognition in science, concluding with a reflection on the implications of this history for current scholarly practices and the ongoing pursuit of more equitable systems of knowledge production.

I. The Texture of Invisible Work: Archival Evidence of Uncredited Contributions

The scientific enterprise between 1890 and 1939 relied heavily on a wide array of tasks that extended far beyond the idealized moment of discovery or the final act of authorship. Archival records, though often fragmented and oblique, reveal women performing intellectually demanding and technically skilled work crucial to the advancement of science, yet systematically excluded from formal credit structures. Examining the nature of these tasks challenges simplistic notions of women acting merely as passive "helpers" and underscores the cognitive labor

involved.

Characterizing Uncredited Tasks

The spectrum of uncredited work performed by women was diverse and required specific competencies:

- **Calculation & Data Processing:** In fields heavily reliant on quantitative data, particularly astronomy and physics, women often served as "computers." This term, predating electronic machines, referred to individuals performing laborious, complex calculations necessary to process raw observational data into usable scientific results. The work at the Harvard College Observatory under Edward C. Pickering is a well-documented example, where dozens of women processed astronomical data captured on photographic plates. While sometimes characterized as routine, these calculations demanded precision, mathematical aptitude, and an understanding of the scientific context. The demanding nature of this work often contrasted sharply with the low pay and lack of recognition afforded to these women, reflecting a tendency to feminize and devalue repetitive yet skilled computational labor.
- **Observation & Recording:** Women, especially those in domestic scientific partnerships as wives or daughters, frequently participated directly in the observational aspects of research. While historical precedents exist, such as Caroline Herschel's astronomical work alongside her brother William, the period 1890-1939 saw continued instances of such collaboration. Margaret Huggins, for example, worked closely with her husband William Huggins in astronomical spectroscopy, actively participating in night-time observations, managing instrumentation, and meticulously recording data in laboratory notebooks. Such involvement blurred the lines between domestic space and scientific workspace, often leading to the woman's contributions being subsumed under her husband's name or familial role.
- **Collections Management & Curation:** Disciplines like botany, zoology, and geology depended fundamentally on the collection, organization, preservation, and preparation of vast numbers of specimens. Women frequently undertook this critical curatorial and preparatory work within museums, herbaria, and laboratories. This involved not only meticulous cataloging and labeling but also specialized skills in preservation techniques, mounting, and dissection – tasks essential for taxonomic study, anatomical research, and building the stable material archives upon which these sciences rested. The skill inherent in careful specimen handling and preparation often went unacknowledged, dismissed as manual or menial despite its necessity for research integrity. Women also contributed as collectors and illustrators, further expanding these collections.
- **Editing, Indexing, & Communication:** The dissemination of scientific knowledge relied on significant textual labor. Women often engaged in editing manuscripts for clarity and accuracy, proofreading galleys, compiling complex indices for books and journals, translating foreign scientific papers, and managing extensive scientific correspondence. Evidence for these activities appears in personal letters discussing manuscript revisions, in financial ledgers noting payments for indexing, or within the acknowledgments sections of published works, where translation work might be mentioned, often framed as spousal devotion rather than skilled linguistic and scientific labor. These tasks were vital for scholarly communication and the consolidation of knowledge, yet rarely conferred authorship or formal credit.

Archival Traces and Their Interpretation

The evidence for these contributions is scattered across diverse archival sources. Formal publications, the primary locus of scientific credit, often remain silent. Instead, researchers must turn to personal correspondence, laboratory notebooks, diaries, institutional administrative records, financial ledgers, and the often-brief acknowledgments or footnotes in published works. Institutional records, for instance, might list women on payrolls in roles designated as "assistant," "secretary," or "technician," documenting their presence and labor within the institution but providing no link to specific research outputs or publications.

The language used in these sources is itself revealing. Women contributors are often referred to by first names,

familial roles ("my wife," "my daughter"), or generic titles like "assistant" or "amanuensis." This terminology reflects and reinforces a gendered hierarchy, positioning their work as supportive and secondary, distinct from the implicitly male-coded role of the primary researcher or author. Deciphering these traces requires careful reading against the grain, recognizing that the absence of formal credit does not equate to an absence of contribution.

The *variety* and *technical nature* of the tasks documented—calculation, observation, specimen handling, editing, indexing, translation—fundamentally challenge any assumption that women's involvement was limited to unskilled "help." These activities demanded specific expertise, precision, and often significant domain knowledge. Labeling this diverse work simply as "assistance" or "support" obscures the cognitive and technical labor involved and reflects gendered assumptions about skill. Furthermore, the *location* of the labor profoundly shaped its visibility. Work conducted within the domestic sphere, often by wives or daughters, was particularly susceptible to being conflated with expected familial duties and thus rendered invisible in formal records. While institutional employment might leave a trace on payrolls, household labor often lacked even this minimal administrative footprint, its documentation reliant on personal papers where it might be framed through affective rather than professional language. The domestic sphere thus functioned as a powerful mechanism for obscuring female scientific labor. Consequently, the *fragmentation* of evidence across disparate source types—letters offering informal details, ledgers providing financial or administrative data, acknowledgments presenting a curated, public recognition—necessitates a multi-sited archival approach. Relying solely on published papers or any single type of record provides an incomplete picture; only by triangulating evidence from these varied sources can a more nuanced understanding of women's uncredited contributions emerge, validating the methodological approach required for this historical recovery.

II. Reconstructing Influence Without Authorship

Given the systemic omission of women from formal authorship lists, recovering their influence requires methods that look beyond traditional markers of credit. Analyzing subtle textual cues in acknowledgments and footnotes, alongside the application of computational techniques like Social Network Analysis (SNA), allows for the reconstruction of involvement and impact that operated outside conventional channels.

Citation Ghosting and the Ambiguity of Acknowledgment

Acknowledgments sections in publications from this period represent a complex site for analyzing credit. While seemingly offering recognition, the language employed often served to simultaneously acknowledge and contain women's contributions. Phrases like "valuable assistance," "tireless support," or thanks for "handling correspondence" could obscure significant labor, positioning it as secondary, non-intellectual, or stemming from personal loyalty rather than professional collaboration. The brevity of such mentions frequently stood in stark contrast to the potential scale of the work involved, as detailed in the previous section.

The practice of acknowledgment was not uniform; variations existed depending on individual authors, institutions, and disciplinary norms. However, gendered patterns are discernible, with women's contributions more likely to be framed in terms of support or assistance compared to male collaborators mentioned for specific intellectual input. Acknowledgments thus functioned as a social technology for managing the credit economy, drawing boundaries around what constituted "real" scientific work worthy of authorship and reinforcing gendered hierarchies by framing women's labor as auxiliary. They performed crucial boundary work, defining the limits of intellectual contribution.

Beyond formal acknowledgments, other textual traces sometimes offer less curated evidence. Footnotes might mention specific calculations performed or data provided by an assistant. Financial records or ledgers can reveal payments for specific tasks like indexing or translation, or document the consistent presence of individuals within a lab or project through payroll entries. These administrative or financial traces, while lacking the narrative framing of acknowledgments, can provide concrete evidence of involvement and labor. Interpreting the *absence* of

mention—silence in author lists, acknowledgments, or institutional reports where participation might be expected—also forms part of this analysis, requiring careful triangulation with other sources and adherence to principles of gentle inference.

Social Network Analysis (SNA) for Invisible Actors

Computational methods, particularly SNA, offer powerful tools for visualizing and analyzing the relational structures within scientific communities, potentially revealing the influence of individuals marginalized in textual records. By mapping connections derived from sources like correspondence networks (tracking who wrote to whom), laboratory rosters or membership lists (showing co-presence in specific institutional contexts), SNA can generate network graphs representing the flow of information and collaboration.

Within these networks, individuals who appear frequently in letters or on rosters but lack corresponding publication records can be identified. SNA metrics, such as centrality measures (e.g., degree centrality, indicating the number of connections; betweenness centrality, indicating a role as a bridge between others), can highlight individuals occupying structurally important positions. Women performing uncredited labor might appear as such 'ghost nodes' – individuals central to communication or workflow but absent from the formal credit system. For instance, a woman managing a prominent scientist's correspondence might exhibit high centrality in a letter network, indicating a crucial role in information dissemination, even if she never authored a paper.

Furthermore, SNA can identify clusters of activity or specific projects where uncredited women were consistently involved alongside credited researchers. Analyzing these clusters can suggest focused, sustained contributions rather than diffuse or occasional help, providing quantitative support for qualitative archival findings. This approach moves beyond anecdotal evidence to reveal systemic patterns of collaboration and dependence on uncredited labor.

Table 1: Illustrative Summary of Potential SNA Findings

The following table provides hypothetical examples based on the types of findings SNA might yield, illustrating how network data can suggest uncredited roles:

Network Analyzed	Potential 'Ghost Node(s)' (Identified via Initials/Role)	Key Connected Credited Scientists	Nature of Connection (Network Metric/Evidence)	Evidence Source(s)	Potential Role/Contribution (Inferred)
Correspondence Network: Pickering's Group (Harvard)	E.L. (Assistant)	E.C. Pickering, W. Fleming	High degree centrality in correspondence related to data requests		Data Management/Dissemination
Lab Roster Connections: Marine Biological Lab (1905-1915)	M.B. (Technician/Preparator)	T.H. Morgan, E.B. Wilson	Consistent co-presence on rosters; mentions in lab supply ledgers		Specimen Preparation, Lab Logistics
Society Membership & Correspondence: Botanical Society	A.R. (Translator/Indexer)	J. Smith, K. Müller	Bridging node (high betweenness) connecting English/German		Translation, Network Facilitation

X			correspondents; Ledger entry		
Domestic Lab Correspondence: Huggins Observatory	M.H. (Spouse/Collaborator)	W. Huggins	High frequency of co-mention in observational logs/correspondence with peers		Observation, Data Recording, Editing

Note: This table is illustrative, based on plausible scenarios suggested by the research context and methods described.

This table demonstrates how SNA can condense complex relational data into a format that highlights potential uncredited actors and their structural positions within scientific networks. It juxtaposes these individuals with credited scientists, specifies the nature of their connection as revealed by network analysis or supporting documents, and suggests possible roles based on contextual evidence, while carefully noting the inferred nature of these roles.

The combination of qualitative analysis of textual traces (acknowledgments, letters, ledgers) and quantitative SNA offers a more robust methodology for reconstructing influence than either approach alone. Qualitative reading provides the necessary context to interpret the connections and patterns revealed by SNA, helping to understand the *nature* of the relationships. Conversely, SNA can quantify and visualize relational patterns suggested by qualitative evidence, demonstrating, for example, that a person mentioned briefly in one source was actually structurally central to a wider network. This synergy allows for stronger, more carefully grounded inferences about the roles and significance of individuals operating outside formal credit structures, revealing structural importance that may contradict their textual marginalization.

III. Gendered Infrastructures: Recognizing Care and Support as Foundational Labor

To fully grasp the significance of women's uncredited contributions, it is necessary to reframe the very nature of their work. Rather than viewing roles like calculator, curator, editor, or specimen handler as merely auxiliary "support," this section argues for understanding them as essential *scientific infrastructure*. Drawing on feminist theories of infrastructure, which emphasize the critical importance of maintenance, care, coordination, and often invisible labor in sustaining systems, this perspective highlights how feminized work formed the foundational bedrock upon which much scientific discovery in this period rested.

Theorizing Support as Infrastructure

Traditional histories of science often focus on moments of breakthrough or theoretical innovation, overlooking the vast amount of labor required to make such moments possible. Scientific infrastructure is not limited to buildings and instruments; it encompasses the systems, processes, and labor involved in managing data, maintaining collections, facilitating communication, and preparing materials for analysis and dissemination. When these crucial tasks are disproportionately performed by women and simultaneously devalued, they constitute a *gendered infrastructure* – essential yet obscured by prevailing assumptions about gender, skill, and intellectual contribution.

Analyzing Specific Infrastructural Roles

Examining specific roles through this lens reveals their infrastructural importance:

- **Curators & Preparators:** The meticulous work of organizing, cataloging, preserving, and preparing biological, geological, or archaeological specimens was fundamental to disciplines reliant on collections. This labor created and maintained the stable material base – the archives of nature or culture – necessary for taxonomic classification, comparative anatomy, evolutionary studies, and countless other research programs. The skills involved, from delicate preparation techniques to systematic organization, were indispensable, yet the work was often performed by women in low-paid, low-status positions within museums and universities, rendered invisible behind the research ostensibly based upon these collections.
- **'Computers' & Calculators:** The large groups of women employed as "computers," particularly in astronomy, functioned as a powerful computational infrastructure. They collectively processed enormous volumes of raw observational data, transforming it through repetitive but complex calculations into tables, charts, and catalogs that formed the basis for theoretical advances and new discoveries. This human infrastructure performed the data processing essential for the scientific output of major institutions like the Harvard College Observatory. The characterization of this work as routine, coupled with low wages, exemplifies how essential infrastructural labor can be devalued when associated with feminized attributes like patience and meticulousness.
- **Secretaries, Typists, Editors:** Administrative and textual labor played a crucial role in managing the flow of information, ensuring the quality and consistency of published work, and maintaining institutional memory. Preparing manuscripts for publication, managing correspondence, translating key texts, and creating indices constituted a vital communication and administrative infrastructure. While women performing these tasks might appear in institutional records, their work was typically segregated into 'support' categories, conceptually separating it from the intellectual work of research and writing, despite its necessity for scientific communication and progress.

Gender Ideology and the Devaluation of Infrastructural Labor

The systematic undervaluation of this infrastructural work was deeply intertwined with the prevailing gender ideologies of the late 19th and early 20th centuries. Women were often stereotyped as naturally suited for tasks requiring patience, neatness, attention to detail, and service – attributes readily mapped onto roles like calculation, curation, and editing. Conversely, traits associated with intellectual leadership, theoretical insight, and discovery were predominantly coded as masculine. This ideological framework facilitated the assignment of essential infrastructural tasks to women while simultaneously justifying lower pay, precarious employment conditions, or reliance on unpaid domestic labor. The economic precarity often associated with these roles was thus directly linked to their gendered nature and perceived lack of intellectual status, creating a self-reinforcing cycle of dependency and invisibility. Because the work was coded feminine, it was deemed less valuable, justifying poor compensation, which in turn reinforced its low status.

Spaces of Infrastructural Labor

The physical and social spaces where this work typically occurred further contributed to its invisibility. Unlike the prominent laboratory bench or the university lecture hall, feminized infrastructural labor often took place in back offices, museum basements, dedicated (and sometimes segregated) computing rooms, or within the private sphere of the household study. This spatial segregation physically reinforced the conceptual separation between the "front stage" of recognized scientific discovery and the "back stage" of essential maintenance and support work. Historical accounts focused on the former naturally tended to overlook the latter, cementing the invisibility of the gendered infrastructure that made the visible work possible. Reframing these roles as infrastructure thus shifts the analytical focus from individual "helpers" to a systemic reliance on labor made cheap or invisible by gender norms, revealing a structural dependency critical to understanding the practice of science in this era.

IV. Variations Across Scientific Landscapes: A Trans-Contextual

Analysis

While the phenomenon of uncredited female labor was pervasive, its specific manifestations varied across different scientific disciplines and institutional settings during the 1890-1939 period. Examining these variations reveals how the broader structure of gendered scientific practice adapted to specific research needs and organizational contexts, while simultaneously highlighting the persistent underlying patterns of exclusion and devaluation.

Disciplinary Differences in Uncredited Labor

The nature of scientific work in different fields shaped the types of uncredited contributions most commonly performed by women:

- **Astronomy:** This field, with its heavy reliance on positional measurements and photographic data analysis, saw the most formalized and large-scale employment of women as "computers". Observatories like Harvard's became centers for this type of work, processing vast quantities of observational data. Women also served as observational assistants, sometimes within family collaborations.
- **Botany, Zoology, and Natural History:** These disciplines depended heavily on collections. Consequently, women's uncredited or under-credited labor often involved specimen curation, preparation, cataloging, and illustration. Field assistance, particularly in the context of expeditions (including colonial settings), and detailed illustration work were also significant areas of contribution.
- **Physics and Chemistry:** Within university or industrial laboratories, women's contributions might involve intricate calculations, meticulous data recording from experiments, maintenance and calibration of sensitive instruments, or preparation of chemical solutions. The increasing complexity and quantitative nature of these fields created needs analogous to those in astronomy, albeit often on a smaller scale.
- **Social Sciences (Emerging):** As fields like sociology, anthropology, and psychology professionalized, women contributed through tasks such as gathering data (conducting surveys, interviews, ethnographic observation), performing statistical calculations, translating source materials, managing archival research, and editing reports. The less formalized laboratory structure in some areas might have made contributions even less visible than in the natural sciences.

Despite these disciplinary variations in the *specific tasks* performed—computation in astronomy, curation and illustration in natural history—the *underlying structure* remained remarkably consistent. Across diverse fields, there was a systematic reliance on women to perform essential labor that was simultaneously deemed non-intellectual or merely supportive, resulting in a lack of formal credit and commensurate status or pay. This consistency suggests the phenomenon was not idiosyncratic to particular disciplines but represented a broader, systemic feature of scientific practice rooted in shared gender ideologies.

Institutional Variations in Visibility and Form

The setting in which scientific work took place also influenced the form and visibility of women's contributions:

- **Universities:** Academic departments often relied on the unpaid labor of professors' wives or daughters for tasks ranging from editing and translation to data analysis and literature reviews, echoing historical patterns of familial collaboration. Universities also employed women in low-paid positions as secretaries, assistants, or technicians, whose contributions to research often went unacknowledged in publications.
- **Museums and Herbaria:** These institutions were significant sites of female employment in roles like preparator, curator, cataloger, and illustrator. While sometimes offering formal employment, these positions were typically low-status and poorly compensated compared to research or directorial roles predominantly held by men.

- **Government Agencies and Observatories:** Large governmental projects or dedicated observatories sometimes created more formalized, albeit segregated, roles for women, such as the astronomical "computers". While providing employment, these roles were often rigidly defined, underpaid, and offered limited opportunities for advancement or recognition as independent researchers.
- **Domestic Labs and Household Science:** A significant amount of scientific work, particularly in the late 19th and early 20th centuries, still occurred within domestic settings. Here, the labor of wives, daughters, or other female relatives was deeply intertwined with domestic duties and familial relationships, making it particularly prone to invisibility and lack of formal recognition.
- **Colonial Stations:** Scientific research in colonial contexts, such as botanical gardens or survey stations, introduced additional layers of complexity. The uncredited labor dynamics likely intersected with colonial power structures and racial hierarchies. Wives of colonial scientists might perform familiar roles, but local or indigenous women might also contribute essential knowledge (e.g., about local flora, fauna, or languages) or labor (e.g., collecting, translating, domestic support) under conditions of even greater invisibility and exploitation, their contributions potentially framed as merely facilitating the 'real' science conducted by Europeans.

These institutional variations shaped the *form* and *traceability* of women's uncredited work. Formal employment in an observatory or museum might leave administrative records, making the labor more visible to historians than work performed within the private domestic sphere. However, this formalization did not necessarily translate into equitable valuation or recognition. Across all these settings, the common thread remains the persistent association of women's labor with support roles and its consequent undervaluation relative to the credited work of predominantly male scientists. The institutional context structured *how* women contributed and *how* their labor was recorded (or not), but the fundamental dynamic of gendered devaluation persisted. Furthermore, the context of colonial science suggests that intersections of gender with race and colonial power likely created distinct conditions of hyper-invisibility and exploitation for non-European women involved in scientific knowledge production, demanding specific attention in further research.

V. Legacies and Modern Parallels: From Historical Erasure to Contemporary Issues

The patterns of uncredited labor and gendered devaluation observed in the scientific networks of 1890-1939 are not merely historical artifacts. They cast long shadows, and understanding their historical roots provides crucial context for analyzing remarkably similar dynamics that persist in contemporary science and technology. Examining these continuities reveals deep-seated structural issues surrounding credit, recognition, and the valuation of different forms of contribution.

Drawing Lines of Continuity

Explicit connections can be drawn between the historical hidden economies of care and present-day challenges:

- **Open Source & Digital Labor:** The historical reliance on often poorly paid or unpaid female "assistants," "calculators," or "secretaries" finds a modern parallel in the digital realm. Open-source software projects, for instance, depend heavily on contributions like code maintenance, documentation, bug reporting, and community management – infrastructural work often performed voluntarily or with little formal recognition, echoing the undervalued support labor of the past. This labor sustains the ecosystem but frequently lacks pathways to formal credit or compensation comparable to core development roles.
- **AI & Data Work:** The large pools of female "computers" meticulously processing data in early 20th-century astronomy bear a striking resemblance to the contemporary global workforce engaged in data annotation, labeling, and cleaning for training artificial intelligence algorithms. This essential work, often outsourced and

performed by a predominantly female workforce in certain regions, is frequently characterized as low-skill, repetitive, and non-technical, justifying low wages and precarious conditions, despite being foundational to AI development. This mirrors the historical gendering and devaluation of computational labor.

- **Research Assistants & Precarious Academic Labor:** The situation of historical assistants, often occupying temporary, poorly paid positions with limited autonomy, resonates with the experiences of contemporary precarious research staff, including postdoctoral researchers, technicians, and research assistants. While acknowledgments might be more common today, contributions related to data management, experimental execution, or lab organization may not always translate into deserved authorship or stable career paths. Women and individuals from marginalized groups remain disproportionately represented in these less secure positions within academia.

The Persistence of Gendered Devaluation

Underlying these parallels is the enduring tendency to devalue certain types of labor based on gendered assumptions. Work associated with care, maintenance, support, communication, or meticulous execution – historically coded feminine – continues to be implicitly or explicitly valued less within many STEM environments than work framed as innovative, theoretical, or demonstrating leadership – historically coded masculine. Tasks like data curation, software maintenance, mentoring, diversity and inclusion work, or community building, while increasingly recognized as important, often struggle for equivalent recognition in promotion, funding, and authorship decisions compared to traditional metrics like first-author publications or large grants.

The boundary between "intellectual contribution" warranting authorship and "technical support" or "care labor" deserving only acknowledgment (or less) remains a site of contestation. This boundary, historically drawn along gendered lines, continues to be negotiated. While contemporary frameworks like the CRediT (Contributor Roles Taxonomy) aim for greater transparency, biases can still influence how these criteria are applied. The fact that roles historically occupied by women (data work, administration, technical support) are often those still debated in authorship discussions suggests the boundary itself carries historical gendered baggage.

Evolution and Ongoing Challenges

Acknowledging continuity does not negate change. There has been progress, including more formalized authorship guidelines, increased awareness of gender bias, and greater visibility for women in many scientific fields. However, challenges persist. Implicit biases can still affect peer review and hiring decisions. Gender pay gaps remain significant in many STEM areas. The "leaky pipeline" phenomenon, where women leave STEM careers at higher rates than men at various stages, points to ongoing systemic issues, including those related to work-life balance, workplace culture, and recognition for diverse contributions.

Understanding the *historical* roots of uncredited labor, as detailed in the preceding sections, is therefore essential for effectively addressing its *contemporary* manifestations. Viewing current issues like the undervaluation of AI data work or the precarity of research assistants not as isolated problems but as modern iterations of long-standing patterns of gendered labor devaluation highlights the need for interventions that address deeply embedded cultural assumptions and power structures within science and technology, rather than relying solely on superficial policy adjustments. The past illuminates the structural nature of present inequities.

VI. Conclusion: Toward a More Inclusive History and Equitable Future

This report has endeavored to recover and analyze the hidden economies of care that sustained scientific networks between 1890 and 1939. The evidence, drawn from diverse archival sources and analyzed through the lenses of feminist historiography, social network analysis, and infrastructural theory, reveals the significant scale,

scope, and essential nature of women's uncredited contributions across various disciplines and institutional settings. From performing complex calculations and managing invaluable collections to editing crucial publications and facilitating communication networks, women performed foundational labor often rendered invisible by prevailing gender ideologies and nascent credit systems focused narrowly on authorship. Recognizing this work not merely as "support" but as integral "gendered infrastructure" allows for a more accurate understanding of how scientific knowledge was actually produced during this critical period of professionalization.

The recovery of these contributions is inherently challenging. The fragmentary nature of the archives, the ambiguity of acknowledgments, and the pervasive silence surrounding much of this labor necessitate careful interpretation and a commitment to gentle inference. The goal is not necessarily to retroactively assign authorship where the historical actors themselves did not – an often impossible and ethically fraught task – but rather to understand the *systems* and *norms* that produced these patterns of exclusion and invisibility. Resisting the temptation to frame these women solely as "lost geniuses" allows for a more nuanced appreciation of the diverse skills and labor involved, while acknowledging the structural constraints that limited their recognition and opportunities.

This historical investigation carries direct implications for contemporary scholarly practice. The processes by which credit was allocated (and withheld) in the past prompt critical reflection on current systems. Modern academic practices – including narrow definitions of authorship, reliance on easily quantifiable metrics like publication counts and impact factors, citation habits that often privilege established figures, and the potential for bias in peer review – risk inadvertently reproducing historical patterns of exclusion. Just as past systems marginalized contributions deemed supportive or infrastructural, often performed by women, current systems may continue to do so if not subjected to ongoing critical scrutiny.

Contemporary citation maps and authorship conventions, despite efforts towards greater transparency, may still silence certain contributions and contributors. Those performing essential work that is less easily captured by traditional authorship criteria – such as data curators, software developers, project managers, technicians, community builders – echo the historical invisibility of women in support roles. Scholars working outside dominant institutions or theoretical paradigms, or those whose contributions are primarily collaborative or infrastructural, may also find their influence underrepresented in standard metrics. The historical analysis presented here underscores how deeply embedded the hierarchy valuing heroic discovery over collaborative or infrastructural labor can be.

Therefore, the act of historical recovery itself becomes intertwined with the ethics of contemporary scholarship. How the history of science is researched and written shapes how contributions are valued in the present. Acknowledging the hidden economies of the past compels a move towards broader conceptions of scholarly contribution, more inclusive and equitable citation practices, and a continued critical examination of how credit is assigned and labor is valued within academia today. Moving towards genuinely equitable credit systems requires more than just procedural adjustments like authorship contribution statements, although these are valuable steps. It necessitates a deeper cultural shift within scientific and academic communities – a shift informed by historical understanding – to actively recognize, value, and reward the full spectrum of labor, care, and expertise that underpins knowledge production. The methodologies employed in uncovering the hidden labor of the past—combining feminist critique, meticulous archival work, network analysis, and infrastructural perspectives—offer potential models not only for historical research but also for analyzing and addressing ongoing inequities in contemporary science and technology. By illuminating the historical roots of exclusion, we gain crucial perspective for building more inclusive and just scientific futures.